

PhD Lecture in Behavioral Asset Pricing: Bounded Rationality and Psychology-free Models

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- **Bounded rationality:** people only get complicated/subtle things wrong
 - Broad theme: we have limited capacity for gathering and processing information
- **Specific topics:**
 - **Limited attention**
 - Underreaction to news
 - Buying vs. selling decisions of individual investors
 - Causal effect of attention on asset prices
 - **Category thinking**
- **Psychology-free models:** we can make interesting predictions without explicit use of psychology
 - Actions of sophisticated traders
 - Actions of noise traders
 - Impacts of investor sentiment (equity and credit market sentiment)

- **Theme:**

- Because of limited attention, investors may underreact to news

- **Underreaction to earnings news**

- Limited attention predicts weaker immediate market reaction and stronger post-earnings announcement drift (PEAD)
- This helps rule out the alternative explanation that the differential PEAD is due to differential information content of earnings surprise

- **Two influential studies:**

- PEAD is stronger for firms that announce earnings at the same time as many other firms (Hirshleifer, Lim, and Teoh, 2009)
- PEAD is stronger for firms that announce earnings on Friday (Della Vigna and Pollet, 2009)

Limited Attention I, ctd.

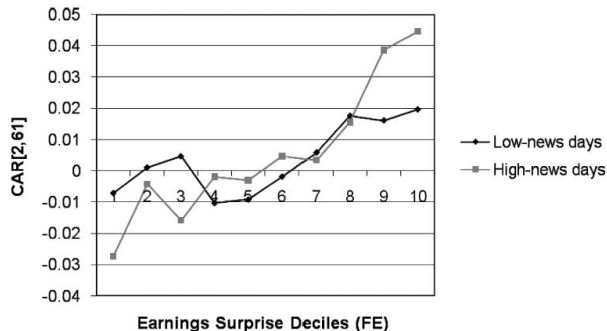
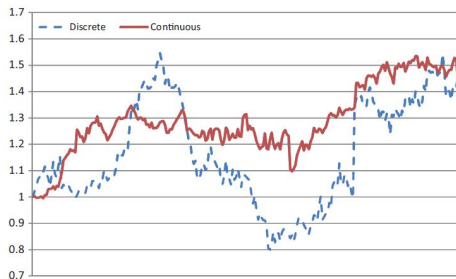


Figure 2. Post-earnings announcement drift: $CAR[2, 61]$. Figure 2 shows the average
Hirshleifer, Lim, and Teoh (2009)

- **A potential explanation for momentum** (Da, Gurn and Warachka, 2014)
 - A series of frequent small changes attracts less attention than infrequent large changes
 - “Frog in the Pan” hypothesis: Investors are more likely inattentive to information arriving continuously in small amounts than that arrives in large amounts in discrete timepoints
 - Momentum is stronger when information arrives continuously as opposed to discretely



Limited Attention I, ctd.

- Underreaction more likely occur when the news is less salient or more subtle
- **Underreaction to demographic information**
 - Della Vigna and Pollet (2008): investors are slow to recognize the impact of demographic shifts on the future profitability of firms with age-sensitive products
 - E.g. Baby boom today predicts higher demand for toys in 6 years so prices of toy-making firms should rise immediately in efficient market
- **Underreaction to technological information**
 - Innovative efficiency: Hirshleifer, Hsu and Li (2013)
 - Track records in innovation ability: Cohen, Diether, and Malloy (2014)
- **Underreaction to the absence of news: No news is news**
 - Giglio and Shue (2014): investors underreact to the probability of merge deal completion contained in the passage of time
 - Hong and Li (2019): investors underreact to the information content when routine insiders stop trading in the months they are “expected” to trade

Limited Attention I, ctd.

- A growing literature documents that investors underreact to news of economically-linked peers (“momentum spillover effect”)
- **Customer-Supplier Links** (Cohen and Frazzini, 2008)
 - Firms are required to report their major customers (>10% of sales)
 - Investors are slow to recognize that good news for a customer is good news for the supplier

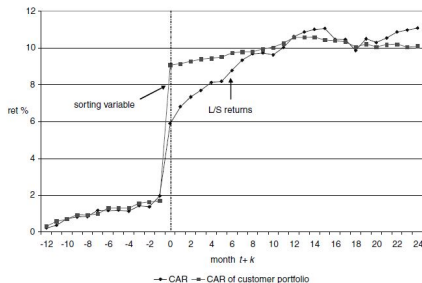


Figure 2. Customer momentum, event-time CAR. This figure shows the average cumulative return in month $t+k$ on a long-short portfolio formed on the firm's customer return in month t . At the beginning of every calendar month, stocks are ranked in ascending order based on the return of a portfolio of its major customers at the end of the previous month. Stocks are assigned to one of five quintile portfolios. The figure shows average cumulative returns (in %) over time of a zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer returns stocks.

Cohen and Frazzini (2008)

Limited Attention I, ctd.

- The same pattern holds for other types of economic linkages
- **Industry and product market peers**
 - Moskowitz and Grinblatt (1999): firms operating in the same industry (industry momentum)
 - Cohen and Lou (2012): Conglomerate firms slower than standalone firms to reflect industry news
 - Huang (2015): Stock prices of multinational firms underreact to foreign industry news
 - Hoberg and Philips (2016): product market peers identified through annual reports
- **Geographic peers**
 - Parsons, Sabbatucci, and Titman (2020): firms headquartered in the same state
 - Jin and Li (2024): firms with overlapped geographic establishments
- **Technological peers**
 - Lee et al. (2018): firms sharing the same type of technologies
- **Analyst co-coverage**
 - Ali and Hirshleifer (2019): firms followed by the same set of analysts

Limited Attention I, ctd.

- Several papers document significant return predictability across debt and equity securities of the same firm
- Han, Subrahmanyam, and Zhou (JFE 2017)
 - Term structure of firms' credit spread predict future firm fundamentals and stock returns
- Addoum and Murfin (RFS 2020)
 - Price of syndicated loans contain useful information for equity prices but investors fail to unravel such information, leading to return predictability from loan market to equity market

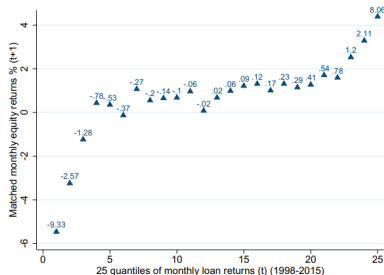


Figure 1

Equity returns versus lagged loan returns

The above plot combines both time-series and cross-sectional relationships between loan returns and linked equity returns in the next month. Loan returns are sorted into twenty-five quantiles over the entire sample period and plotted against the next month's equity returns for the same borrowers. Average loan returns for each quantile are reported next to the triangles.

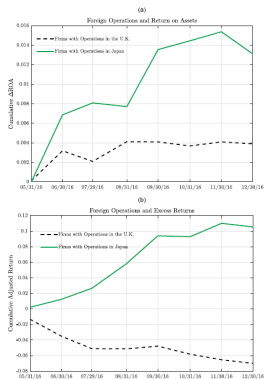
Addoum and Murfin (RFS 2020)

- The source of underreaction could be “market segmentation”
 - Both debt and equity are claims to the same underlying cash flows/asset values
 - Investors trading in debt markets have low overlap with those trading in equity markets, leading to underreaction to cross-asset information
 - Returns to cross-asset momentum are lower among equities held by mutual funds also trading in debt securities

Limited Attention I, ctd.

- Firm value is also affected by shocks to other asset classes, including currency and commodities
- **Cross-asset return predictability**
- Bae, Da and Zurita (2023)
 - Equity investors of U.S. multinationals underreact to firms' currency shocks from countries in which they operate
- Ho and Lauwers (2023)
 - Aggregate positions of traders in the commodity futures market can predict the cross section of stock returns for commodity producers

Figure 1. (Color online) Foreign Currencies and Operations: The Brexit Case



Limited Attention II

• Theme:

- Attention is more important for individual investors' buying decisions than for their selling decisions (Why?)
- Barber and Odean (2008) show that there is indeed stronger buying interest than selling interest, among individual investors, for attention-grabbing stocks
- Stocks with extreme returns, high volume, or news announcement

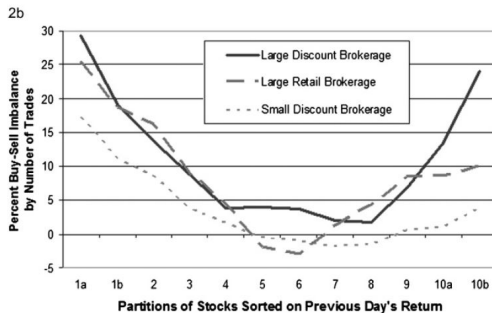


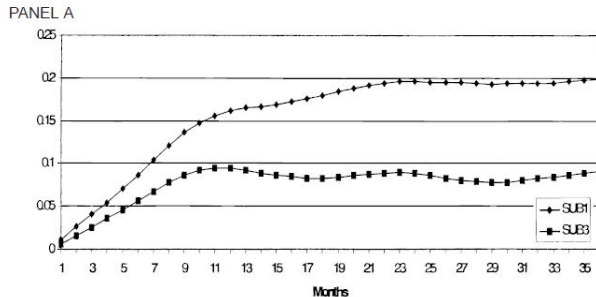
Figure 2
Individual Investor Buy-Sell Imbalances by Number of Trades for Stocks Sorted on the Current Day's Abnormal Trading Volume and Previous Day's Return

Limited Attention III

- Because of limited processing capacity, investors may only pay attention to one kind of information
- **Hong and Stein (1999)**
- Two types of investors:
 - Newswatchers, who only process private information
 - Momentum traders, who only track past prices, and whose demand is a function of the past price change
- Assume that private information diffuses slowly through the newswatcher population
 - In an economy with only newswatchers, this would generate momentum
- In an economy with both kinds of investors, we generate both momentum and long-term reversals
 - It is optimal for momentum traders' demand to be a positive function of the past price change
 - Price eventually overshoot because momentum traders keep buying even after (positive) private information is fully incorporated into prices

Limited Attention III, ctd.

- Hong, Lim, and Stein (2000)
- Test the Hong and Stein (1999) hypothesis that momentum is due to slow infusion of private information
 - Find that momentum is stronger among small stocks and stocks with low analyst coverage



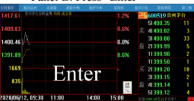
Causal Effect of Attention on Asset Prices

- The key challenge in identifying the causal effect of attention on asset prices is that investor attention is endogenous to stock fundamentals (maybe unobservable)
- Chen, An, Wang and Yu (JF 2023) circumvent this issue by exploring attention spillover effect
- They leverage the unique setting in China where the order of stock displayed on screen is determined by the stock's listing code
- Investors are more likely to pay attention to neighbouring stocks if they have positive investment experience with the focal stock

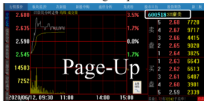
Panel A. Input "GZMT"



Panel B. Press "Enter"



Panel C. Press "Page-Up"



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Panel E. Press "Enter"



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Causal Effect of Attention on Asset Prices

- Because attention leads to excessive buying demand, stocks displayed next to those with higher returns in the past two weeks are associated with higher returns in the future week, which are reverted in the long run

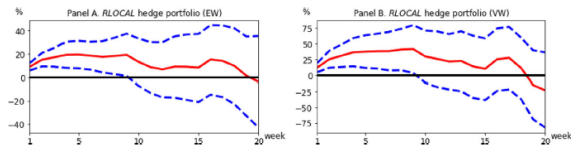


Figure 5. Cumulative return of the long-short portfolio based on *RLOCAL*. This figure shows the annualized cumulative CH4-alpha (in percentage points) to the equal- and value-weighted long-short portfolios (P5-P1) based on *RLOCAL* from week $t + 1$ to week $t + 20$, as well as the 95% confidence interval. For each stock, *LOCAL* is calculated as the value-weighted average return over the past two weeks of the 10 closest stocks on screen display, and *RLOCAL* is the residual of the cross-sectional regression of *LOCAL* on the focal stock's return over the past two weeks. Portfolios are formed at the end of week t based on *RLOCAL*. (Color figure can be viewed at wileyonlinelibrary.com)

Chen, An, Wang and Yu (JF 2023)

Category Thinking

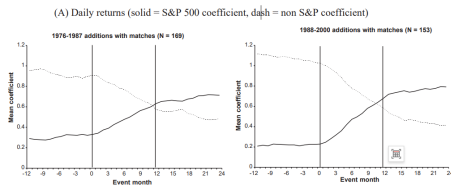
Category Thinking

- A basic feature of the way humans think about a complex world is that we put things into categories and form beliefs at the level of these categories (Rosch and Lloyd, 1978)
- It is plausible that such category-based thinking also applies to investment
 - As there are thousands of individual securities and funds, investors may simplify their decision-making by putting financial assets into categories
 - We often use labels such as “value stocks,” “growth stocks,” “small-cap stocks”
- **Barberis and Shleifer (2003)**
 - Model where some investors put assets into categories and form beliefs about the assets' future returns at the category level
- Novel prediction is **excess return comovement**
 - Returns of assets in the same category comove more than can be explained by fundamentals, driven by correlated demand shocks

- Numerous studies documented excess return comovement among firms or funds sharing similar characteristics, supporting Barberis and Shleifer (2003):
 - Stocks belonging to the same indices (Barberis et al., 2005; Greenwood, 2008; Boyer, 2011)
 - Firms headquartered in the same state (Pirinsky and Wang, 2006)
 - Stocks with similar levels of prices (Green and Hwang, 2009)
 - Firms covered by similar sets of analysts (Israelsen, 2016)
 - Stocks held by a common set of mutual funds (Anton and Polk, 2014)
 - Dividend-paying stocks (Hameed and Xie, 2019)

Excess Return Comovement among Stocks in the Same Index

- Barberis, Shleifer and Wurgler (2005) test the prediction of category thinking that stock that is added to (removed from) the S&P 500 index will experience a large increase (decrease) in its loading on the S&P return



$$R_{j,t} = \alpha_j + \beta_{j,SP500} R_{SP500,t} + \beta_{j,nonSP500} R_{nonSP500,t} + v_{j,t}$$

Psychology-free Approaches

Psychology-free Approaches

- The approaches discussed so far have mostly been rooted in a psychological concept
- Some behavioural finance models shed light on the asset pricing facts without appealing to any specific psychology
- A common structure for such a model is De Long et al. (1990), where less than fully rational investors (“noise traders”) interact with more rational traders known as “arbitrageurs”
- Even without additional psychological assumptions, it yields many testable predictions:
 - Level of mispricing will be higher for assets whose valuation is more uncertain
 - Level of mispricing will be higher for assets whose characteristics make arbitrage more difficult
 - Arbitrageurs’ action in correcting mispricing is informative about future asset return

Actions of Sophisticated Agents

- If some assets are mispriced, the trades of sophisticated investors should predict returns
- **Who are these sophisticated investors?**
- **Firm managers**
 - Key assumption is that firm managers can detect and respond to market mispricing
 - Issuance/repurchase decisions: change the supply of shares in response to investor demand
 - Personal transaction in own firms' stocks — “corporate insider trading”
- **Short sellers**
 - Stocks heavily shorted have low subsequent returns (Asquith, Pathak, and Ritter 2005)
- **Hedge funds**
 - Chen, Da, and Huang (2018): Net arbitrage trading (NAT) of HFs strongly predicts stock returns in the cross-section
- **Analysts?**
 - Lee and So (2017) show that abnormal analyst coverage positively predict subsequent return

Actions of Sophisticated Agents

- Interestingly, actions of sophisticated investors, when aggregated to the market level, can predict market return in the same direction as stock-level return
 - Baker and Wurgler (2000): share of equity issues in total equity and debt issues negatively predict market return
 - Seyhun (1988): aggregate trading by corporate insiders positively predict future market return
 - Rapach, Ringgenberg, and Zhou (2016): aggregate short interests negatively forecast market return

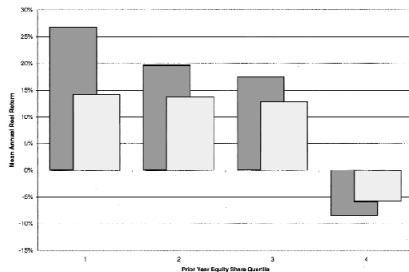


Figure 2. Mean equity returns by prior-year equity share in new issues, 1928–1997. Mean annual real returns on the CRSP value-weighted (light) and equal-weighted (solid) indexes by quartile of the prior-year share of equity issues in total equity and debt issues. Real returns are created using the Consumer Price Index from Ibbotson Associates (1998).

Baker and Wurgler (2000)

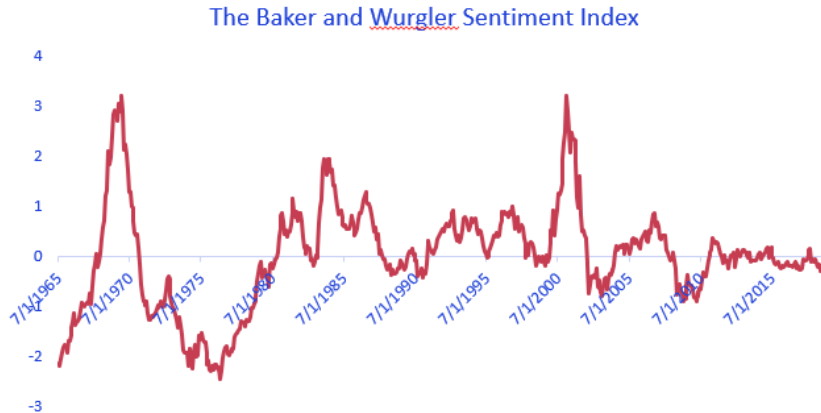
Actions of Noise Traders

- If some assets are mispriced, the trades and actions of unsophisticated agents should negatively predict returns
- **Several studies confirm this:**
- **Frazzini and Lamont (2008)**
 - Stocks held by mutual funds with large inflows in the previous three years have low subsequent returns
- **Barber, Odean, and Zhu (2008) and Hvidkjaer (2008)**
 - Stocks heavily bought by individual investors in the previous year perform poorly in the following year
 - Use small trades in TAQ dataset to identify individuals and the Lee-Ready algorithm to identify buy/sell order
- **Da, Engelberg, and Gao (2011)**
 - Higher investor attention as captured by Google search volume is correlated with abnormal retail buying and leads to elevated stock price initially but lower return over the long-run

Market-based Measures of Sentiment

- Baker and Wurgler (2006)
- Measure investor “sentiment” using the principal component of following six proxies from market data:
 - **Trading volume:** TURN is defined as the natural logarithm of the turnover ratio of trading volume to the number of shares listed on the NYSE, detrended by the 5-year moving average
 - **Closed-end fund discount:** CEFD is defined as the average difference between the net asset values of closed-end stock fund shares and their market prices
 - **IPO activity:** NIPO is defined as the number of IPOs
 - **First-day returns:** RIPO is defined as the average of first-day returns on IPOs
 - **Equity issues:** EIT is defined as the ratio of equity issues to total equity and debt issues by all corporations
 - **Dividend premium:** DP is defined as the difference between the average market-to-book ratios of dividend payers and nonpayers
- Each of those measures has been proposed by prior studies as capturing investor sentiment

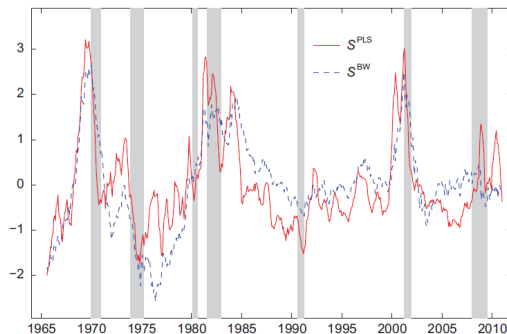
Baker and Wurgler Sentiment Index



Baker and Wurgler Sentiment Index

- Huang, Jiang, Tu and Zhou (2015)
- Refine the BW sentiment index by using the partial least squares (PLS) regression to more efficiently extract the return forecasting information from six underlying proxies
- They show the new index has much greater predictive power than the original sentiment indices have both in and out of sample

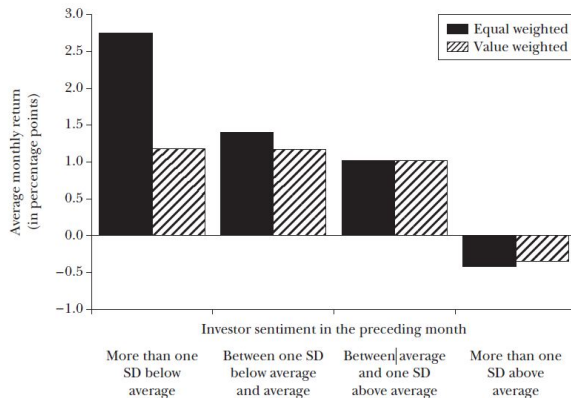
Investor Sentiment Aligned: A Powerful Predictor of Stock Returns



Effects of Investor Sentiment on Stock Markets (I)

- Baker and Wurgler (2007): Sentiment should predict market returns with a negative sign

Figure 6
Sentiment and Market Returns



Effects of Investor Sentiment on Stock Markets (I) — Role of Uncertainty

- Birru and Young (JFE 2022) show that the return predictability of investor sentiment is much greater when uncertainty is high
- Consistent with the idea that sentiment exhibit its strongest effects on asset prices at times when valuations are most subjective

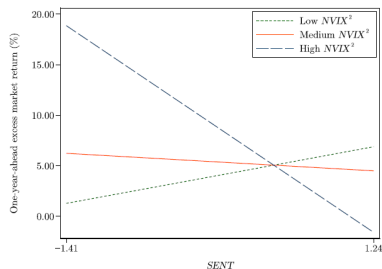
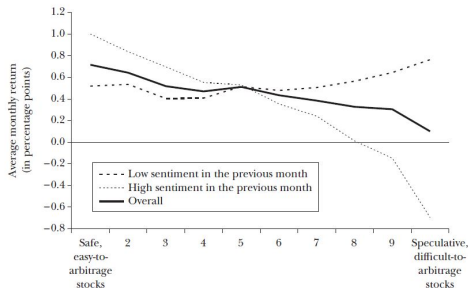


Fig. 1. Sentiment and Market Returns. Panel A: Daily. Panels A and B plot the relation between sentiment and future market returns conditional on low, medium, and high uncertainty at the daily and monthly frequency, respectively. Equation (1) provides the parameter estimates. Low, medium, and high uncertainty correspond to the 10th, 50th, and 90th percentile of the distribution, respectively. In Panel A, VIX is the CBOE volatility index, the x-axis is the range between the 10th and 90th percentile of the Da et al. (2015) sentiment index ($FEARS$), and the y-axis is the one-day ahead excess market return. In Panel B, $NVIX^2$ is the square of the news implied volatility index from Manela and Moreira (2017), the x-axis is the range between the 10th and 90th percentile of the Baker and Wurgler (2006) sentiment index ($SENT$), and the y-axis is the one-year ahead excess market return. The sample period spans July 2004 to December 2011 for Panel A and July 1965 to March 2016 for Panel B.

Effects of Investor Sentiment on Stock Market (II) — Cross-Sectional Implication

- When market-wide sentiment is high, stocks that are hard to arbitrage and difficult to value will be overpriced and have lower subsequent returns
- Baker and Wurgler (2006) confirm this prediction
 - Identify hard-to-arbitrage and difficult-to-value stocks as small stocks, young firms, volatile stocks, unprofitable stocks, and non-dividend paying stocks

Figure 5
Sentiment and Future Returns



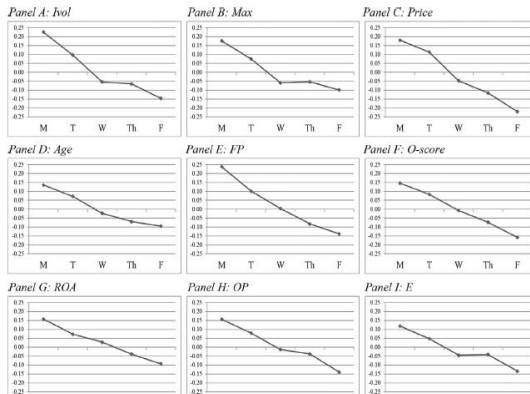
Effects of Investor Sentiment on Stock Market (III) — Anomalies

- Stambaugh, Yu, and Yuan (JFE 2012) argue that anomalies should be amplified following high sentiment
 - The effect of sentiment should be stronger for overpricing than for underpricing due to short-sale constraints

Anomaly	Long leg			Short leg			Long-short		
	High sentiment	Low sentiment	High - low	High sentiment	Low sentiment	High - low	High sentiment	Low sentiment	High - low
Failure probability	0.77 (2.16)	1.14 (3.74)	-0.38 (-0.81)	-1.10 (-1.54)	1.25 (2.26)	-2.34 (-2.60)	1.86 (3.25)	-0.10 (-0.24)	1.96 (2.72)
Ohlson's O (distress)	0.42 (1.14)	0.61 (2.06)	-0.19 (-0.41)	-0.98 (-1.69)	0.61 (1.33)	-1.59 (-2.15)	1.40 (3.81)	-0.00 (-0.01)	1.40 (2.85)
Net stock issues	0.64 (2.22)	0.75 (3.04)	-0.11 (-0.28)	-0.50 (-1.26)	0.63 (2.10)	-1.13 (-2.28)	1.14 (5.71)	0.12 (0.88)	1.02 (4.20)
Composite equity issues	0.53 (1.93)	0.72 (3.08)	-0.19 (-0.52)	-0.28 (-0.72)	0.69 (2.13)	-0.97 (-1.91)	0.81 (3.19)	0.02 (0.13)	0.79 (2.46)
Total accruals	0.37 (0.82)	1.07 (3.10)	-0.71 (-1.25)	-0.57 (-1.06)	0.84 (2.22)	-1.41 (-2.14)	0.94 (3.11)	0.23 (1.04)	0.70 (1.88)
Net operating assets	0.50 (1.36)	0.92 (3.01)	-0.43 (-0.90)	-0.57 (-1.37)	0.69 (2.20)	-1.26 (-2.41)	1.07 (4.66)	0.24 (1.29)	0.83 (2.84)
Momentum	0.78 (1.69)	1.43 (4.12)	-0.64 (-1.11)	-1.24 (-2.14)	0.34 (0.76)	-1.58 (-2.16)	2.03 (4.49)	1.09 (3.12)	0.93 (1.64)
Gross profitability	0.59 (1.84)	0.79 (2.73)	-0.20 (-0.47)	-0.06 (-0.18)	0.64 (2.48)	-0.70 (-1.62)	0.65 (2.93)	0.15 (0.64)	0.50 (1.53)
Asset growth	0.79 (2.14)	1.22 (3.26)	-0.43 (-0.81)	-0.60 (-1.30)	0.68 (1.92)	-1.27 (-2.20)	1.39 (5.04)	0.54 (2.34)	0.85 (2.37)
Return on assets	0.61 (1.60)	0.66 (2.10)	-0.05 (-0.10)	-1.10 (-1.78)	0.44 (1.00)	-1.55 (-2.02)	1.72 (4.01)	0.22 (0.65)	1.50 (2.74)
Investment-to-assets	0.44 (1.19)	1.38 (4.13)	-0.94 (-1.90)	-0.47 (-1.14)	0.78 (2.25)	-1.25 (-2.32)	0.91 (4.48)	0.60 (2.93)	0.30 (1.06)
Combination	0.56 (1.72)	0.95 (3.51)	-0.39 (-0.93)	-0.68 (-1.54)	0.65 (1.96)	-1.32 (-2.41)	1.23 (6.64)	0.31 (2.64)	0.93 (4.25)

Effects of Investor Sentiment on Stock Market (III) — Anomalies

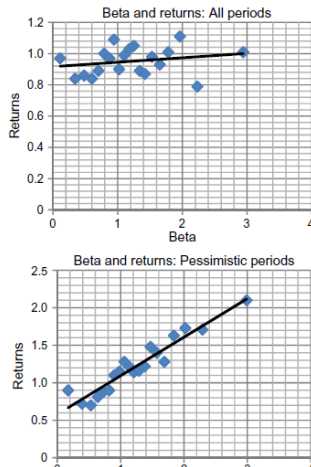
- Birru (JFE 2018) documents that long-short anomaly returns are strongly related to the day of the week
 - In particular, long-short strategies earn highest return on Monday and lowest return on Friday when the short-leg is more speculative
 - Consistent with psychology literature that mood increases (decreases) on Friday (Monday)



Effects of Investor Sentiment on Stock Market (IV) — Risk-Return Trade-off

- Shen, Yu, and Zhao (JME 2017) argue that sentiment also affects cross-sectional risk-return trade-off
 - When sentiment is high, sentiment-driven investors tend to require a smaller compensation for the risk they bear
 - In addition, riskier stocks may be more subject to sentiment shifts and become more overpriced
 - Coupled with short-sale constraints, the risk-return trade-off may be attenuated or even reversed during high sentiment period
- Similar evidence is documented by Antoniou, Doukas and Subrahmanyam (2016) for the security market line

Figure 1 (Color online) Beta, Sentiment, and Returns

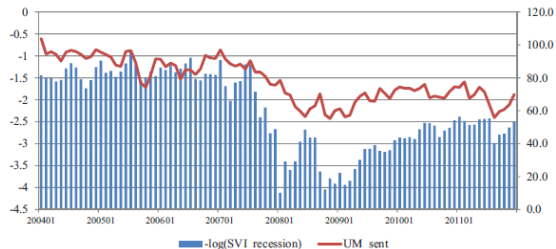


Effects of Investor Sentiment on Stock Market (IV) — Risk-Return Trade-off

- Chen et al. (RFS 2025) examine the characteristics-beta-sorted portfolios (e.g., momentum beta)
 - Different from anomaly portfolios
- High-beta portfolios yield higher returns than low-beta portfolios post high-sentiment periods
- This suggests that characteristic-based factor beta is more likely a proxy for the level of mispricing, rather than exposure to systematic risks
- Why? Recall the effect of sentiment on SML and macro risk premium is opposite

Other Measures of Aggregate Sentiment — Google Search

- Da, Engelberg, and Gao (RFS 2014)
- Aggregate the volume of Google search queries related to household concerns and construct a Financial and Economic Attitudes Revealed by Search (FEARS) index as a new measure of investor sentiment
- Benefits of FEARS:
 - More transparent
 - Higher frequency — Daily
 - Reveal attitudes rather than inquire about them
- Gao, Ren and Zhang (2020) document similar evidence for international markets



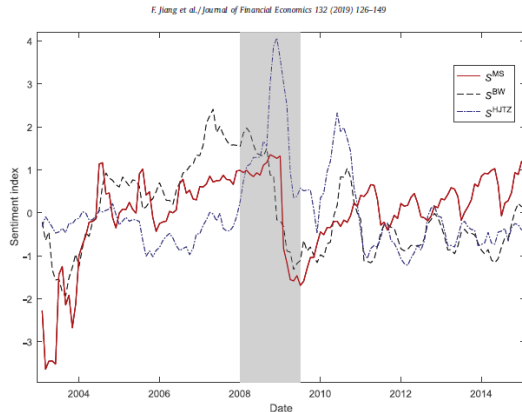
FEARS and asset return:

FEARS is negatively related with contemporaneous return,
positively with future return

FEARS predicts temporary increase in volatility
FEARS predicts mutual fund flows out of equity funds to
bond funds

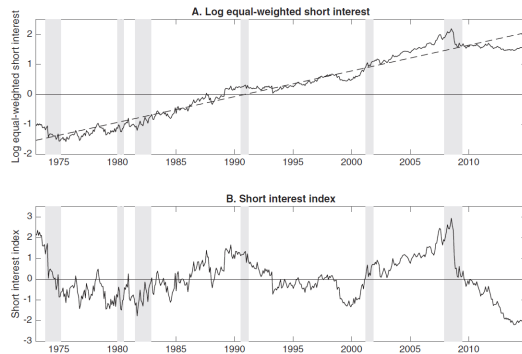
Other Measures of Aggregate Sentiment — Manager Sentiment

- Jiang et al. (JFE 2019)
 - Manager sentiment index based on the aggregated textual tone in 10-Ks, 10-Qs, and conference call transcripts
 - Manager sentiment is a strong negative predictor of future aggregate stock market returns



Other Measures of Aggregate Sentiment — Aggregate Short Interests

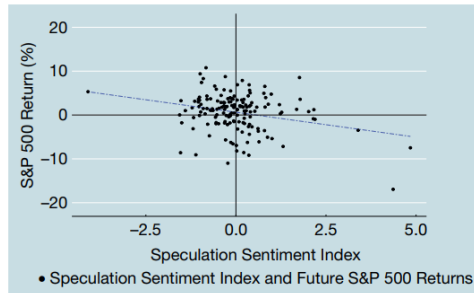
- Short sellers are sophisticated investors who can correct mispricing induced by sentiment traders
- Short interests aggregated to market level is a strong contrarian predictor of future market return



Other Measures of Aggregate Sentiment — Flows to Leveraged ETFs

- Leveraged ETFs provide amplified gains/losses to investors who bet on the direction of future market movement
- Heavily traded and cater to speculative investors
- Hence flows to such ETFs mainly reflect speculative investor's demand
- Davies (2022) constructs a Speculative Sentiment Index (SSI) based on this insight
 - SSI is the difference between share change in the leveraged-long ETFs and share change in the leveraged-short ETFs
- 1-standard-deviation increase in the monthly index is associated with a 1.14%–1.67% decline in broad market stock in the next month

Graph C. SSI Versus S&P 500 r_{t+1}



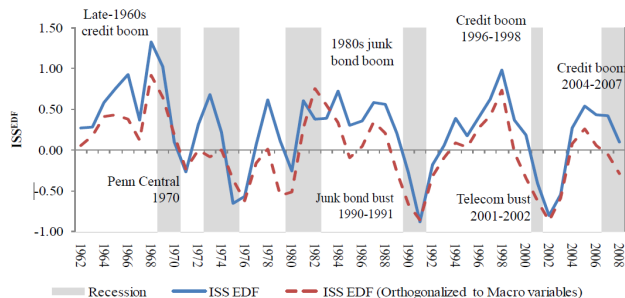
Measuring Investor Mood/Emotions

- Edmans et al. (2022): Music sentiment
- The positivity of songs that individuals choose to listen to
- **Advantages:**
 - Real-time, continuous measure of national sentiment
 - Language free and comparable globally
 - Exogeneous to financial market events
- **Main findings:**
 - Music sentiment is positively correlated with same-week equity market returns and negatively correlated with next-week returns
 - Music sentiment also predicts increases in net mutual fund flows, and precedes a rise in stock market volatility
 - Negatively associated with government bond returns, consistent with a flight to safety

Credit Market Sentiment

Credit Market Sentiment

- Greenwood and Hanson (RFS 2013) construct a sentiment measure for the credit market
 - The idea is to compare the credit quality of high issuance firms with that of low issuance firms
 - The insight is that a significant decline in issuers' credit quality is a reliable signal of credit market overheating



Credit Market Sentiment Forecasts Excess Corporate Bond Returns

- Economic magnitudes are significant:
 - $1\text{-}\sigma$ increase in ISS EDF $\Rightarrow$ cumulative excess returns fall by 7.30 %-points over the following 2 years

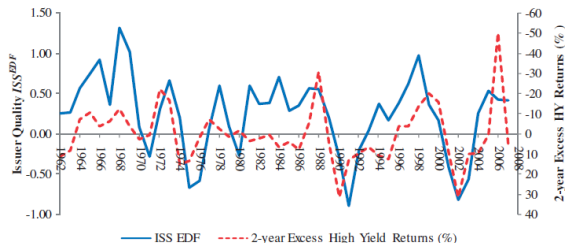


Figure 3
Issuer quality and subsequent high-yield excess returns

Issuer quality (left axis) plotted alongside cumulative excess high-yield bond returns for the following two years (right axis). Returns are plotted in reverse scale, so the negative correlation appears positive visually. Issuer quality is measured with ISS^{EDF} , the difference between the average EDF decile of high- and low-debt issuers from 1962–2008.

Greenwood and Hanson (RFS 2013)

Credit Market Sentiment and Economic Growth

- Credit market overheating is associated with a higher probability of a future financial crisis and economic downturns
 - Schularick and Taylor (2012); Lopez-Salido, Stein, and Zakrajsek (2017); Baron and Xiong (2017)

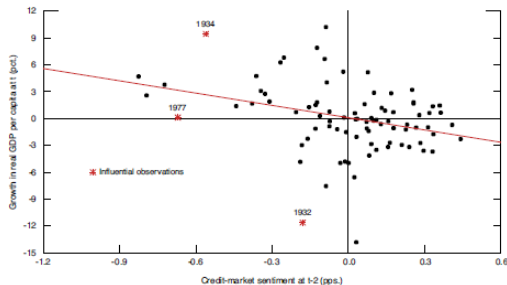


FIGURE II

Credit-Market Sentiment and Economic Growth, 1929–2015

The scatter plot depicts a visual representation of the relationship between credit-market sentiment in year $t - 2$ and the growth in real GDP per capita from year $t - 1$ to year t , implied by the specification in column (1) of Table II. See the text and Figure III for the definition of influential observations

Lopez-Salido, Stein, and Zakrajsek (QJE 2017)

Credit Expansion Predicts Banking Crisis

- Baron and Xiong (QJE 2017) find bank credit expansion predicts increased bank equity crash risk
- Despite the elevated crash risk, credit expansion predicts lower mean bank equity returns in subsequent one to three years
- Bank credit expansion is distinct from equity market sentiment captured by dividend yield
- The result suggests overoptimism and neglect of crash risk by bank equity investors during credit expansions

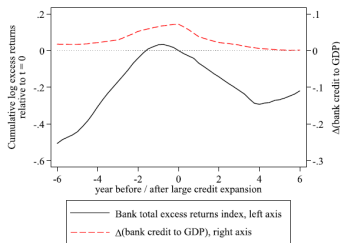


FIGURE II

Bank Equity Prices and Bank Credit before and after Large Credit Expansions

The past three-year change in bank credit to GDP (i.e., $\Delta(\frac{\text{bank credit}}{\text{GDP}})$) and the bank total excess log returns index are plotted before and after a large credit expansion. A large credit expansion is defined as *credit expansion* exceeding the 95th percentile threshold, which is calculated for each country and each point in time using only past information to avoid any future-looking bias. $\Delta(\frac{\text{bank credit}}{\text{GDP}})$ and bank total excess log returns are pooled averages across time and countries, conditional on the given number of years before or after the start of a banking crisis. The average bank log returns are then cumulated from $t = -6$ to $t = +6$, and the level is adjusted to be 0 at $t = 0$. Observations are over the sample of 20 countries, 1920–2012.

Predictable Financial Crisis

- Greenwood, Hanson, Shleifer and Sorensen (JF 2022)
- Using historical data on postwar financial crises around the world
- The combination of rapid credit and asset price growth over the prior three years (R-zone) is associated with a 40% probability of entering a financial crisis within the next three years

Panel A: Business Sector																
	Crisis within One Year				Crisis within Two Years				Crisis within Three Years				Crisis within Four Years			
	(1.1)	(1.2)	(1.3)	(1.4)	(2.1)	(2.2)	(2.3)	(2.4)	(3.1)	(3.2)	(3.3)	(3.4)	(4.1)	(4.2)	(4.3)	(4.4)
<i>High Debt Growth</i> ^{Bus.} (β^h)	6.9** [2.3]		5.3** [2.1]		11.6*** [3.0]		9.5** [2.5]		16.8*** [3.3]		11.5** [2.7]		15.6** [2.7]		10.3* [2.2]	
<i>High Price Growth</i> ^{Bus.} (δ^h)		0.4 [0.1]	−0.4 [−0.2]			4.8 [0.9]	3.8 [0.8]			10.5 [1.4]	7.4 [1.1]			10.7 [1.5]	7.6 [1.2]	
<i>R-Zone</i> ^{Bus.} (γ^h)			5.3 [0.8]	9.0 [1.1]			7.8 [1.3]	17.9* [2.1]			19.4** [2.8]	33.7*** [3.3]			19.4** [2.6]	33.0** [3.1]
Sum of coefficients ($\beta^h + \delta^h + \gamma^h$)	6.9	0.4	10.2	9.0	11.6	4.8	21.1	17.9	16.8	10.5	38.2	33.7	15.6	10.7	37.3	33.0
<i>t</i> -Statistic ($\beta^h + \delta^h + \gamma^h$)			1.2				2.1				3.2				3.1	
<i>R</i> ² (<i>within</i>)	1.6	0.0	1.9	1.1	2.5	0.7	3.6	2.3	3.8	2.4	7.8	6.1	2.8	2.1	6.2	4.8
<i>N</i>	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258

Credit Market vs. Equity Market Sentiment

- Empirical measures of credit market sentiment are only modestly correlated with measures of equity market sentiment
 - Suggests we should not think of equity and credit markets as fully integrated
- Measures of expected returns on the stock market, such as Shiller's CAPE, do not predict future GDP growth (Lopez-Salido et al., 2017)
 - Suggests credit market sentiment has potentially greater macroeconomic significance

Determinants of Credit Market Sentiment

- Short-term rates are low or have fallen in recent past
 - Suggestive of “reaching for yield”
- Term and credit spread is low or has fallen in the recent past
- Past default rates have been low or falling
- Investors flow into high-yield bonds (Ben-Rephael, Choi and Goldstein, 2021)
- Intermediary balance sheets are strong
- Evidence supports frictional/behavioral interpretations

Early Indicator of the Credit Boom

- Ben-Rephael, Choi and Goldstein (JFE 2021)
- Early indicator of credit boom based on intra-family flow shifts towards high-yield bond mutual funds
- Intra-family flow likely capture investors' deliberate asset allocation decision (rather than saving/investment decision)
- It leads other indicators of credit boom such as growth in financial intermediary balance sheets, increase in shares of high-yield bond issuers, and downturns of various measures of credit spreads
- Attributes the origins of the credit boom to an increase in investors' demand for HY bonds, perhaps due to biased perceptions of default risk

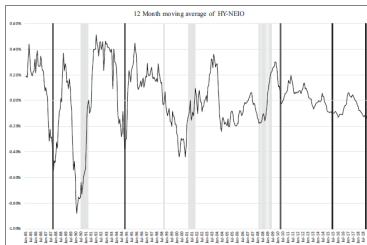


Fig. 1. 12-month moving averages of HY-NEO. In this figure we plot the 12-month moving averages of HY-NEO from January 1985 (i.e., February 84-January 85) through December 2018. HY-NEO is net exchanges (exchange-in minus exchange-out) from high-yield corporate bond funds normalized by the end-of-previous-month assets. The data are obtained from the ICI and run from February 1984 through December 2018. The three light gray columns represent NBER recession periods. The dark gray columns represent the 1987 market crash, the Mexican Peso crisis in 1994, the European sovereign debt crisis in early 2010, the high-yield spread hike during June 2015–February 2016, and the market crash in the last quarter of 2018.

A Behavioral Account of the Credit Cycle

- Narrative of credit cycle by Minsky and Kindleberger (*Manias, Panics, and Crashes: A History of Financial Crises*)
- Full model is developed in Bordalo et al. (2018) and Greenwood, Hanson, and Jin (2024)
- Key feature of the model is **reflexivity**: the feedback loop between investors' biased perceptions and market outcomes
- **Main idea:**
 - Investors hold extrapolative beliefs based on borrowers' recent default history
 - During a boom, investors extend credit on attractive terms because borrowers have paid back in the past
 - Even if borrowers' fundamental position deteriorates, this may not be directly visible for a while because borrowers roll over debt at low rates
 - When the reckoning eventually comes, markets overreact, slowing the recovery

- **Bounded rationality:**

- Limited attention leads to underreaction to news
- Category thinking leads to excess comovement

- **Psychology-free approaches:**

- Actions of sophisticated traders predict returns
- Actions of noise traders negatively predict returns
- Investor sentiment affects market returns and cross-sectional patterns

- **Credit market sentiment:**

- Distinct from equity market sentiment
- Predicts financial crises and economic downturns
- Behavioral models explain credit cycle dynamics